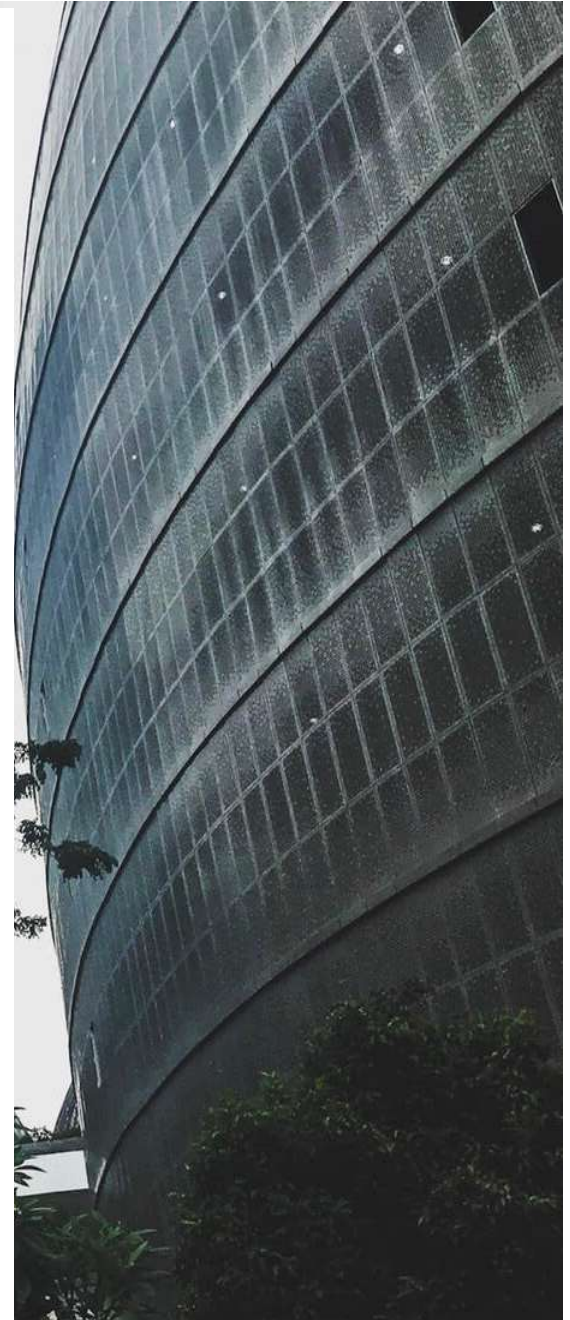


COURSE MATERIAL

**IF670 – Mobile Cross Platform Programming
S1 INFORMATICS GRADUATE PROGRAM
FACULTY OF ENGINEERING AND INFORMATICS**

**INFORMATICS STUDY PROGRAM
FACULTY OF ENGINEERING AND INFORMATION
UNIVERSITAS MULTIMEDIA NUSANTARA**

Gedung B Lantai 5, Kampus UMN
Jl. Scientia Boulevard, Gading Serpong, Tangerang, Banten-15811 Indonesia
Telp: +62-21.5422.0808 (ext. 1803), email: lab.fti@umn.ac.id, web: umn.ac.id



MODUL 9

(Camera, Storage, Filesystem)

THEME DESCRIPTION

Mahasiswa akan mempelajari cara menghandle camera, storage, dan filesystem.

WEEKLY LEARNING GOAL (SUB-GOAL LEARNING)

Mahasiswa mampu memanfaatkan fitur native seperti kamera, penyimpanan, dan sistem file dalam pengembangan aplikasi.

SOFTWARE / HARDWARE REQUIREMENTS

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

PRACTICUM STEPS

1. Create a new project and install the required libraries:

`npx expo install expo-camera expo-media-library expo-image-picker`

2. Add the following configuration to your `app.json` file in your Expo project..

```
"plugins": [
  "expo-router",
  [
    "expo-splash-screen",
    {
      "image": "./assets/images/splash-icon.png",
      "imageWidth": 200,
      "resizeMode": "contain",
      "backgroundColor": "#ffffff",
      "dark": {
        "backgroundColor": "#000000"
      }
    }
  ],
  [
    "expo-camera",
    {
      "cameraPermission": "Allow $(PRODUCT_NAME) to access your camera"
    }
  ],
  [
    "expo-media-library",
    {
      "photosPermission": "Allow $(PRODUCT_NAME) to access your photos",
      "savePhotosPermission": "Allow $(PRODUCT_NAME) to save photos to your library",
      "isAccessMediaLocationEnabled": true
    }
  ]
],
```

3. Modify your `index.tsx` file to use the **camera** and access the **gallery**.

```
import { Text, View, Button, Image, StyleSheet } from "react-native";
import { useState } from "react";
import * as ImagePicker from "expo-image-picker";
import { Camera } from "expo-camera";

export default function Index() {
  const [image, setImage] = useState<string | null>(null);

  // 📷 OPEN CAMERA
  const openCamera = async () => {
    const permission = await Camera.requestCameraPermissionsAsync();

    if (!permission.granted) {
      alert("Camera permission is required!");
      return;
    }

    const result = await ImagePicker.launchCameraAsync({
      mediaTypes: ImagePicker.MediaTypeOptions.Images,
      quality: 1,
    });

    if (!result.canceled) {
      setImage(result.assets[0].uri);
    }
  };

  // 🖼️ OPEN GALLERY
  const openGallery = async () => {
    const permission = await ImagePicker.requestMediaLibraryPermissionsAsync();

    if (!permission.granted) {
      alert("Gallery permission is required!");
      return;
    }

    const result = await ImagePicker.launchImageLibraryAsync({
      mediaTypes: ImagePicker.MediaTypeOptions.Images,
      quality: 1,
    });

    if (!result.canceled) {
      setImage(result.assets[0].uri);
    }
  };
};
```

```
return (
  <View style={styles.container}>
    <Text style={styles.text}>
      Name - NIM
    </Text>

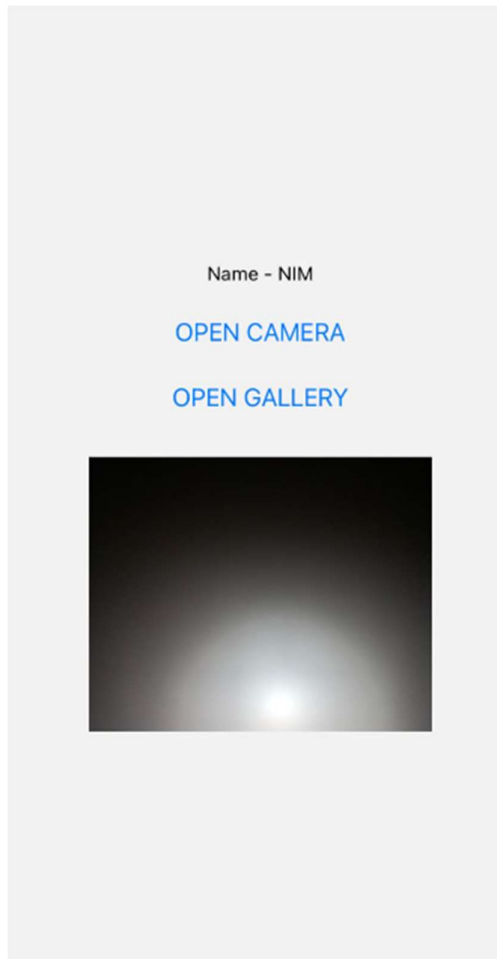
    <View style={styles.button}>
      <Button title="OPEN CAMERA" onPress={openCamera} />
    </View>

    <View style={styles.button}>
      <Button title="OPEN GALLERY" onPress={openGallery} />
    </View>

    {image && (
      <Image source={{ uri: image }} style={styles.image} />
    )}
  </View>
);
}
```

```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    alignItems: "center",
    justifyContent: "center",
  },
  text: {
    marginBottom: 10,
  },
  button: {
    marginVertical: 5,
    width: 150,
  },
  image: {
    width: 250,
    height: 200,
    marginTop: 20,
  },
});
```

4. Below is the result of the implementation of the code above.



6. After successfully implementing the camera, install an additional supporting library:
`npx expo install expo-file-system`

ASSIGNMENT

Modify the tutorial you have completed:

Create a function and a button labeled "SAVE IMAGE" using the expo-file-system library that you installed earlier. Then verify whether the image has been successfully saved to your device gallery.

REFERENCE